

30 CHAPTERWISE IMPORTANT QUESTIONS

Mathematics Standard (041) | CBSE 2026-27

Areas Related to Circles

CBSE Mathematics 2026-27 | Board-examinable

20 numerical / problem-solving questions and 10 theory, proof or method questions. Workings and answers are written for Class X revision.

Question 1

Find the area of a 90 degree sector of radius 7 cm. Use $\pi = 22/7$.

Answer:

$$\text{Area} = \frac{90}{360} \times \pi \times r^2 = 38.5 \text{ cm}^2.$$

Question 2

Find the area of a 90 degree sector of radius 14 cm. Use $\pi = 22/7$.

Answer:

$$\text{Area} = \frac{90}{360} \times \pi \times r^2 = 154 \text{ cm}^2.$$

Question 3

Find the area of a 60 degree sector of radius 21 cm. Use $\pi = 22/7$.

Answer:

$$\text{Area} = \frac{60}{360} \times \pi \times r^2 = 231 \text{ cm}^2.$$

Question 4

Find the area of a 180 degree sector of radius 10 cm. Use $\pi = 22/7$.

Answer:

$$\text{Area} = \frac{180}{360} \times \pi \times r^2 = 157.143 \text{ cm}^2.$$

Question 5

Find the area of a 45 degree sector of radius 14 cm. Use $\pi = 22/7$.

Answer:

$$\text{Area} = \frac{45}{360} \times \pi \times r^2 = 77 \text{ cm}^2.$$

Question 6

Find the area of a 120 degree sector of radius 7 cm. Use $\pi = 22/7$.

Answer:

$$\text{Area} = \frac{120}{360} \times \pi \times r^2 = 51.3333 \text{ cm}^2.$$

Question 7

Find the area of a 30 degree sector of radius 21 cm. Use $\pi = 22/7$.

Answer:

$$\text{Area} = \frac{30}{360} \times \pi \times r^2 = 115.5 \text{ cm}^2.$$

Question 8

Find the area of a 90 degree sector of radius 28 cm. Use $\pi = 22/7$.

Answer:

$$\text{Area} = \frac{90}{360} \times \pi \times r^2 = 616 \text{ cm}^2.$$

Question 9

Find the area of a 60 degree sector of radius 14 cm. Use $\pi = 22/7$.

Answer:

$$\text{Area} = \frac{60}{360} \times \pi \times r^2 = 102.667 \text{ cm}^2.$$

Areas Related to Circles - Continued

Question 10

Find the area of a 72 degree sector of radius 35 cm. Use $\pi = 22/7$.

Answer:

$$\text{Area} = \frac{72}{360} \times \pi \times r^2 = 770 \text{ cm}^2.$$

Question 11

Find the arc length of a 60 degree sector of radius 7 cm. Use $\pi = 22/7$.

Answer:

$$\text{Arc length} = \frac{60}{360} \times 2\pi r = 7.33333 \text{ cm.}$$

Question 12

Find the arc length of a 90 degree sector of radius 14 cm. Use $\pi = 22/7$.

Answer:

$$\text{Arc length} = \frac{90}{360} \times 2\pi r = 22 \text{ cm.}$$

Question 13

Find the arc length of a 120 degree sector of radius 21 cm. Use $\pi = 22/7$.

Answer:

$$\text{Arc length} = \frac{120}{360} \times 2\pi r = 44 \text{ cm.}$$

Question 14

Find the arc length of a 180 degree sector of radius 10 cm. Use $\pi = 22/7$.

Answer:

$$\text{Arc length} = \frac{180}{360} \times 2\pi r = 31.4286 \text{ cm.}$$

Question 15

Find the arc length of a 45 degree sector of radius 14 cm. Use $\pi = 22/7$.

Answer:

$$\text{Arc length} = \frac{45}{360} \times 2\pi r = 11 \text{ cm.}$$

Question 16

Find the arc length of a 30 degree sector of radius 28 cm. Use $\pi = 22/7$.

Answer:

$$\text{Arc length} = \frac{30}{360} \times 2\pi r = 14.6667 \text{ cm.}$$

Question 17

Find the arc length of a 180 degree sector of radius 7 cm. Use $\pi = 22/7$.

Answer:

$$\text{Arc length} = \frac{180}{360} \times 2\pi r = 22 \text{ cm.}$$

Question 18

Find the arc length of a 60 degree sector of radius 21 cm. Use $\pi = 22/7$.

Answer:

$$\text{Arc length} = \frac{60}{360} \times 2\pi r = 22 \text{ cm.}$$

Question 19

Find the arc length of a 90 degree sector of radius 35 cm. Use $\pi = 22/7$.

Answer:

$$\text{Arc length} = \frac{90}{360} \times 2\pi r = 55 \text{ cm.}$$

Areas Related to Circles - Continued

Question 20

Find the arc length of a 120 degree sector of radius 14 cm. Use $\pi = \frac{22}{7}$.

Answer:

Arc length = $\frac{120}{360} \times 2\pi r = 29.3333$ cm.

Question 21

State the main formula, rule or theorem used in Areas Related to Circles.

Answer:

Sector area = $\frac{\theta}{360} \times \pi r^2$; arc length = $\frac{\theta}{360} \times 2\pi r$.

Question 22

What do the important symbols or quantities in Areas Related to Circles represent?

Answer:

θ is the central angle and r is the radius.

Question 23

Write the correct method for solving a Areas Related to Circles numerical.

Answer:

Use the given angle as the fraction $\frac{\theta}{360}$ of the full circle, then calculate with correct square units for area.

Question 24

Which unit, sign convention or condition must be checked in a Areas Related to Circles answer?

Answer:

Write the given data first, use the stated convention, and include units wherever the final quantity has a unit.

Question 25

Why must the final answer in Areas Related to Circles be checked against the given condition?

Answer:

Checking catches wrong signs, arithmetic errors and values that are impossible in the original situation.

Question 26

What is one common error in Areas Related to Circles questions?

Answer:

Students often substitute a value in the wrong formula or skip a condition. Write the formula before substituting.

Question 27

How should full working be shown in a Areas Related to Circles numerical?

Answer:

Show the formula or theorem, substitution, simplification, calculation and final answer in that order.

Question 28

When is a diagram, table or graph useful in Areas Related to Circles?

Answer:

Use it when it makes the given quantities and their relationship clear before calculation.

Areas Related to Circles - Continued

Question 29

How can a student decide which formula to use in Areas Related to Circles?

Answer:

Identify the quantity asked, list the known quantities, then select the formula or theorem connecting them.

Question 30

What makes a high-scoring answer in Areas Related to Circles?

Answer:

Correct reasoning, complete steps, an appropriate formula, a clearly stated final result and a quick verification.

Sources: [CBSE Mathematics curriculum 2026-27](#); [CBSE Class X sample papers and marking schemes 2026-27](#); [NCERT textbooks](#).

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